

LITTLE BLUE BOOK NO.

Edited by E. Haldeman-Julius

860

Our Insect Enemies

Vance Randolph

Drawings by Peter Quinn

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ENEMIES ***

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Edited by E. Haldeman-Julius

Our Insect Enemies

Vance Randolph

Drawings by Peter Quinn

HALDEMAN-JULIUS COMPANY

GIRARD, KANSAS

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OUR INSECT ENEMIES

HOUSEFLIES AND THEIR KINDRED

The common housefly (*Musca domestica*) is perhaps more thoroughly domesticated than any other insect; it lives and flourishes wherever man has established his settlements, and does not thrive elsewhere. It is mentioned by many of the ancient writers, and has evidently been associated with man from the remotest antiquity, doubtless adapting itself gradually to his changing modes of life. It still breeds in the refuse which accumulates about his buildings, and still invades his home to partake of his food. When one considers the number of flies which get into modern dwellings despite glass and wire and modern insecticides, it is difficult to understand how our sturdy fathers managed to survive the summers in mediaeval Europe. Picture Thomas Aquinas battling with swarms of flies for his food! The housefly's favorite breeding place has always been in horse manure, and man has until recent years had great quantities of this nutritious substance about him; with the coming of the automobile this condition has been changed somewhat, but even the cleanest of American cities still affords sufficient decaying matter to assure a bountiful crop of flies. At some future day perhaps we shall have cities so spotless that there will be nowhere a speck of garbage large enough to nourish a maggot, but that day is still to come, and it doth not yet appear what we shall be.

Meanwhile, as Herrick says, the housefly is with us always. "They are present from early spring to late fall, even remaining far into the winter. They are troublesome in kitchens and dining rooms because of their abundance, their proneness to get into food, and their generally filthy habits. Until recent years the housefly has been generally regarded as somewhat of a scavenger and has been considered of value to humanity because of its aid in the removal of wastes that are a menace to human welfare. The eggs of the housefly are often deposited on decaying vegetable matter that is allowed to accumulate in the vicinity of human habitations and the maggots that hatch from the eggs live on this decaying matter and aid in destroying it. Thus it must be conceded, perhaps, that houseflies do assist somewhat in the removal of foul and dangerous waste matters and, to this extent, are of benefit. On the other hand, it has been conclusively

shown that this modicum of benefit is greatly overbalanced by their role in disseminating dangerous diseases. It has been shown that houseflies carry the germs of cholera, typhoid fever, cholera infantum, and tropical dysentery on their feet, legs, bodies and in their digestive tracts. There can be no doubt of the responsibility of the housefly for much sickness and many deaths.”

Although the adult housefly is equally fond of fresh and putrid matter, flitting from filth to food and back again, the eggs are always laid in decaying substances, preferably horse manure. In the absence of this delicacy the eggs may be deposited in human excrement, or even cow manure if it is not too dry. A single female fly often deposits one hundred and fifty eggs at a single sitting, and lays as many as six or seven hundred in the course of a season.

The egg is white and shaped like a grain of wheat; it measures about one-twentieth of an inch in length, and hatches in from ten to twenty-four hours, the period of incubation varying with the temperature. The second or larval stage is represented by a wormlike creature called a maggot, which crawls about and devours great quantities of nourishing manure. When the maggot is about five or six days old it has attained a length of nearly three-eighths of an inch, and transforms into the pupa; it is no longer active, ceases to take food, and the outer skin turns into a hard, brown, dry case called a puparium. After five or six days of rest the puparium is burst open and the adult fly crawls out.

The newly emerged flies mate within two or three days, and the young females are soon laying their first batch of eggs. Thus the whole development may occur in about fourteen days, and Herrick thinks that in Washington, D. C., it may be complete in ten days or less, according to the temperature, with ten or twelve generations in a summer. “One can hardly realize,” says Herrick, “the enormous numbers that such rapid development is capable of producing. Inside of two months, one female fly can give rise to many millions of progeny. For the purpose of illustration, we will assume that a female fly lays 100 eggs. If these hatch and all the larvae come to maturity, about one-half will probably be males and the other half females. Then at the end of the first generation there will be fifty egg-laying females. At this rate, at the end of the eighth generation there would be produced about 1,875,000,000,000 adults. Of course, in nature, a very large part of

these would die before reaching maturity, so that actually one female would probably never produce such an enormous number of individuals. However, even under normal conditions tremendous numbers are produced.”

The adult housefly is unhappily too familiar to require a lengthy description. The total length is about one-fourth of an inch, there are two membranous wings and six hairy legs, while the thorax is gray with four dark longitudinal stripes above. Besides the hairs and bristles on the body and legs, which entangle great numbers of germs, the sticky *pulvilli* of the feet gather bacteria as if they had been especially developed for that purpose. Thus every housefly scatters bacteria broadcast wherever he goes; one has only to allow a fly to walk across a plate of sterile gelatin to see large colonies of bacteria produced in every foot-print. Esten and Mason's examination of some 400 specimens showed that the number of bacteria carried ranged from 550 to 6,600,000 per fly, with an average of nearly a million and a quarter.

The general impression is that the housefly does not travel any great distance from its birth-place, but there is considerable doubt about this matter. Hewitt has taken them flying at least eighty feet above the ground, and it is obvious that small insects which rise so high may be carried long distances by the wind. Copeman, Howlett and Merriman have attacked the problem by marking large numbers of specimens, releasing them, and then watching for them in traps set at various distances; they recovered several which had travelled more than three-fourths of a mile. Dr. Hodge found plenty of houseflies on the cribs of the Cleveland waterworks, which are located six miles out in Lake Erie, and which offer no breeding places for flies. He naturally concluded that they were blown six miles by the wind.

It has been pretty well established that adult flies do not retain and disseminate disease germs which they swallowed as maggots in the manure, but only those which they acquire in the adult state. It has been definitely proved, however, that the bacilli of typhoid fever are carried on the bodies and legs of houseflies, and also that the bacilli are often swallowed and survive the passage through the body, as living bacilli have frequently been demonstrated in flyspecks. The most important practical consideration is, of course, the actual frequency with which houseflies do transfer pathogenic organisms from filth to food intended for human consumption, thus

spreading disease, and this is in its very nature a difficult matter to investigate.

Typhoid fever is a disease affecting the intestines, and the bacilli which cause it frequently remain in the body for long periods after the patient has apparently recovered. The germs are passed out in the urine and fecal matter. As Herrick puts it: "If the excreta or urine containing these bacilli are deposited where they are accessible to flies, for instance in open privies, the chances are high that the bacilli will be carried on the bodies of these insects back to the kitchens and dining rooms and be deposited on our food. During the Spanish-American War, flies were traced by their whitened feet from the lime-sprinkled, open latrines or privies to the dining tables of the soldiers in camp. It makes one shudder to think of the thousands of open closets in the small towns of the United States to which flies have access and in which they breed and from which they may come direct to our kitchens and dining rooms."

Excluding the greater variability of the water supply, the fact that typhoid is so much more prevalent in country districts than in the city may be due almost entirely to the abundance of flies, since other opportunities for infection are no greater than in the city. Certain South American cities have very good water, but typhoid rages nevertheless, because the natives expose their excreta within a few feet of the street markets, and myriads of flies carry the bacilli from the dunghill to the food. "In our own country," says Brues, "the seasonal incidence of typhoid fever corresponds to some extent with fly prevalence, and still more significant is its greater summer prevalence in regions where systems for sewage disposal are not generally installed.... The greater uniformity throughout the year in New York, where the opportunities for fly-borne infection are curtailed, is very marked. Another way in which the housefly can aid in the spread of typhoid is through infecting milk on dairy farms where carriers are present and offer the flies a chance to become infected."

The vessels in an ordinary country dairy are nearly always more or less accessible to flies, and warm milk is an excellent medium in which to cultivate bacteria. When a fly which has been feeding on the excreta of a typhoid patient falls into a can of milk, there is a very good chance that the disease may be spread all along the milkman's route. Dr. Taylor, of the Colorado State Board of Health, reports a case of this sort: "In the city of

Denver we had a very sad as well as a plain demonstration of the transmission of typhoid fever by flies and milk. Early in August of this year the wife of a dairyman was taken with typhoid fever, remaining at home about three weeks before her removal to the hospital, August 28. During the first two weeks of September we received reports of numerous cases of typhoid fever in the northern portion of Denver, and upon investigation found that all these cases had been securing their milk from this dairy.

“An inspection of the dairy was then made, and in addition to learning of the illness of the dairyman’s wife, we also found the dairyman himself suffering with a mild case of typhoid, but still up and delivering milk. The water supply of the dairy was fairly good. However, we found that the stools of both the wife and husband had been deposited in an open privy vault located only thirty-five feet from the milk house, which was unscreened and open to flies. The gelatin cultures exposed for thirty minutes in the rear of the privy vault and in the milk house among the milk cans gave numerous colonies of typhoid bacilli, as well as colon bacilli and the ordinary germ-life. The source of infection in the dairyman’s wife’s case is unknown, but I am positive that in all the cases which occurred on this milk route the infection was due to bacilli carried from this vault by flies and deposited upon the milk cans, separator and utensils in the milk house, thereby contaminating the milk. The dairyman supplied milk to 143 customers. Fifty-five cases of typhoid fever occurred and three deaths resulted therefrom.”

As long ago as 1873 Dr. Nichols noted that the number of cholera cases on his ship seemed to vary directly with the abundance of flies. Later experiments demonstrated the presence of the cholera bacillus upon the bodies and in the excreta of flies fed upon cholera-infected material, and it has been shown that flies sometimes transfer the cholera bacilli from human excreta to milk. Dr. Tsuzuki, of the Japanese Army Medical Service, found the germs of cholera upon flies taken in cholera-ridden houses, and demonstrated that flies carry the organisms from place to place. Macrae, commenting on an outbreak of cholera in a jail in India, remarked that the place was infested with “a plague of flies ... which were present in swarms when the disease broke out, and it was an observation of daily experience to see them settling on cholera stools whenever possible.” Dr. Macrae goes on to say that the flies undoubtedly carried the cholera germs to the prisoners’

food, and adds: "The practical lesson is, that flies should be looked upon in the light of poisonous agencies of the worst kind during cholera epidemics, and it is clear that if they find access to poison they will carry and distribute it, and every possible means should be taken to prevent their getting into contact with either food or drink of any kind, and to those having to deal with large bodies of men it is a lesson more easily learnt than put into practice."

Because of the flies' liking for expectorated saliva, it has long been suspected that tuberculosis may be carried about by these insects. Several investigators have carried out experiments in which flies have been fed upon the sputa of consumptives, and the tuberculosis bacilli have been found in their intestines and fly specks. Frederick T. Lord, after a long series of laboratory investigations, concluded that: "(1) Flies may ingest tubercular sputum and excrete tubercle bacilli, the virulence of which may last for at least fifteen days. (2) The danger of human infection from tubercular fly specks is by the ingestion of the specks on food. Spontaneous liberation of tubercle bacilli from fly specks is unlikely. If mechanically disturbed, infection of the surrounding air may occur." Dr. Lord goes on to suggest that "tubercular material (sputum, pus from discharging sinuses, fecal matter from patients with intestinal tuberculosis, etc.) should be carefully protected from flies, lest they act as disseminators of the tubercle bacilli. During the fly season greater attention should be paid to the screening of rooms and hospital wards containing patients with tuberculosis and laboratories where tubercular material is examined. As these precautions would not eliminate fly infection by patients at large, foodstuffs should be protected from flies which may already have ingested tubercular material."

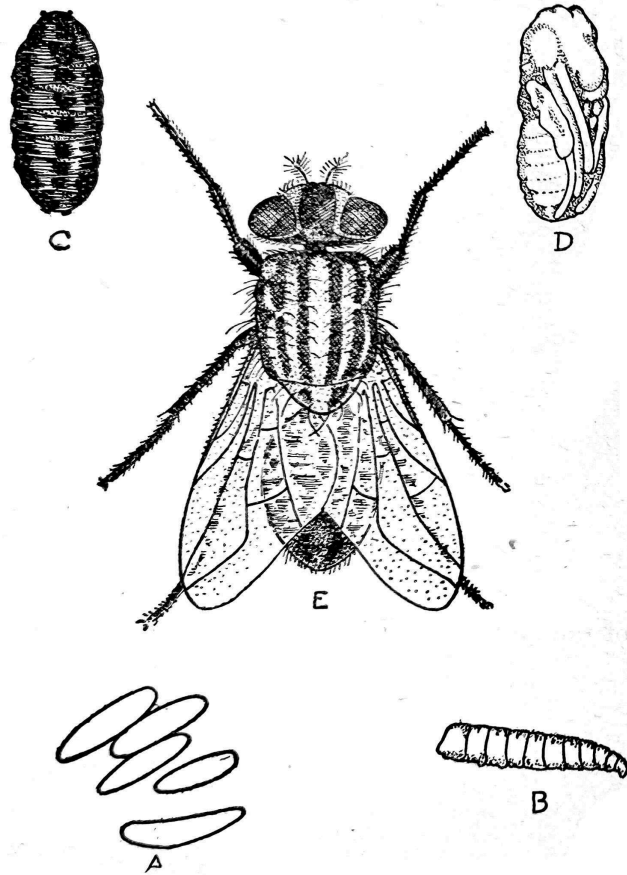


Fig. I. The Metamorphosis of the Housefly (*Musca domestica*). A, eggs, considerably enlarged; B, young maggot; C, puparium; D, pupa; E, adult.

Numerous British army surgeons long ago hazarded the guess that flies carry dysentery, but the first clear case is that of an epidemic which occurred in the Worcester State Hospital in 1910. Houseflies were especially abundant at the time, and Dr. Orton, after a painstaking experimental study of the affair, concluded that flies were entirely responsible for the epidemic.

There is now a fairly general agreement that flies spread the organisms which cause the infantile diarrhea known as summer complaint, which kills more infants than any other single cause except lack of sufficient and proper food. Fraser, Nash and others have noted that the more flies the more cases of summer complaint, and Sandilands explains the immunity of the children

of the rich by the smaller number of flies in their houses. It is certainly true that the mortality of bottle-fed infants is very much higher than of breast-fed babies, but whether Jackson is justified in attributing this mainly to the infection of cow's milk by flies is a question.

Lucian Howe has long contended that purulent ophthalmia—the sore-eye so common up and down the valley of the Nile—is distributed mainly by the housefly. The disease is always most prevalent where the flies are most abundant, and in flyless desert regions there is practically no ophthalmia. The natives are singularly dirty and indolent, and flies are often seen to settle undisturbed about a pair of badly infected eyes. Dr. Howe has captured a number of these flies, and found that their feet were covered with the same bacilli found in the secretion of the inflamed conjunctiva.

Braun, Demetriades and others report that gonorrheal and other eye infections are often carried by houseflies. Welanders, in 1896, according to Howard, “observed an interesting case where an old bed-ridden woman in a hospital became infected. It seems that her bed was along side that of another patient who had blennorrhoea, but that a screen which did not reach the ceiling separated the beds. Thus all means of infection except through the agency of flies was apparently absent. The investigator found that flies bore living gonococci upon their feet three hours after they had been soiled with secretion, since they infected sterilized plates with which they came in contact.”

Nuttall and Jepson, after reviewing the entire literature of the subject, think that “the evidence regarding the spread of Egyptian ophthalmia by flies appears to be conclusive, and the possibility of gonorrheal secretions being conveyed by flies cannot be denied.”

Bubonic plague is now definitely known to be spread by fleas which are parasitic on rats, but many observers still cling to the view that the housefly also acts as a distributor. It is certainly true that flies are killed by feeding upon the bodies of persons who die of the plague, and Yersin, at Hong Kong in 1894, showed that these dead flies contain active plague bacilli. Whether the fly plays any serious part in the actual spread of bubonic plague is still a mooted question.

The housefly has plenty of natural enemies, but they do not seem to reduce the number to any material extent. One of the most destructive is a

minute fungus which grows inside the fly's body until it finally causes death. The dead flies one finds on window panes in the fall, surrounded by a whitish ring, are the victims of these fungus growths. There are several species of these fungi; in some cases at least the attack begins with a spore which attaches itself to the outside of the body, and grows a long thread-like root which enters the body through one of the spiracles or breathing pores.

Several species of protozoan parasites have been found in the intestines of flies, along with a few minute parasitic worms, but they are not known to cause any great amount of inconvenience. Small red mites are often seen fastened to flies' bodies; some of them are true parasites which suck the juices from the body of their host, and look very much like the chiggers which annoy the higher animals in the Southern States. With other mites, according to Howard, "the flies simply act as aeroplanes to carry the mites from one place to another. A free ride seems to be the only object for which they have attached themselves to the fly."

Spiders, if given an opportunity, destroy a great many flies, but as the webs are usually destroyed by the housekeeper's broom they do very little execution. The little centipede known as the water bug kills a great many flies, but as it works at night gets little credit for its good offices.

Hornets are sometimes seen to capture and carry off houseflies, but it is doubtful if any great number of the pests are killed by these insects. An English entomologist named Westwood once circulated a story to the effect that "the Americans, aware of their service in destroying flies, sometimes suspend a hornet's nest in their parlors." A little later one Benjamin D. Walsh, an American, wrote that "some persons in America have turned the insect-devouring propensity of the hornets to good purpose by suspending one of their nests in a house much infested by the common housefly. In such a situation we have been told that they soon make a clearance of the obnoxious flies; and so long as you do not meddle with them they will not meddle with you." To me these two quotations are the sheerest nonsense; any one who has any acquaintance with hornets will find it difficult to visualize an American farmer intentionally filling his house with hornets for any reason whatsoever. I have never been able to find anybody who has tried out this method; if there are any such among my readers I should like very much to hear from them. Please address me in care of the publisher of this booklet.

The logical way to combat the housefly would be to destroy the eggs and maggots as we do in the case of the mosquito, but it is not practical at present. In rural districts, where the dung of livestock is always lying about, and the human excreta exposed in shallow open closets, very little progress has been made. Of course, the danger of typhoid may be avoided by keeping the excreta of typhoid patients where flies cannot get at it, and the flies certainly could not spread tuberculosis if no tubercular sputum was exposed. These facts are taken advantage of in the cities, where there are plenty of flies, but very little typhoid, because the water closet system is such that flies cannot get access to the excreta. In communities where there is no adequate sewage disposal and where flies are plentiful, the individual can only screen them out of his own dwelling as best he can. Howard estimates that more than \$10,000,000 are spent every year for screen wire in the United States alone.

Many ingenious devices for destroying flies inside houses are on the market—poisons, sticky fly paper, fly traps and fly swatters. One of the most useful of these is sticky fly paper cut in strips and suspended from the ceiling. This will often clear a room of flies when the ordinary sheets of the same paper lying flat on a table are quite inefficacious. In Japanese hospitals, says Howard, “they take a whole potato and stick it full of toothpicks, put fly paste on the toothpicks, and hang the potatoes from the ceiling over the patient’s bed on a cord. The flies all gather on the potato, and when it is full they throw the potato away and make a new trap. The toothpicks are placed about one-fourth of an inch apart, and the potato presents the appearance of a porcupine.”

C. F. Hodge is all for building large traps and placing them out of doors in early Spring, arguing that a great many newly emerged flies are thus caught before they reproduce, and that it is better to catch one fly in the Spring than thousands in the Autumn. Brues even says that “practical traps whereby fly-larvae in stored manure may be caught and destroyed before transformation have also been devised.”

Some progress has been made in the treatment of manure piles with poisons. Howard found that either chloride of lime or kerosene kill the maggots all right, but the cost of both is prohibitive. Davis worked with iron sulphate and reported that manure could be cleaned of maggots and deodorized at a cost of about two cents per horse per day. Howard, Hens

and others have advocated fly-tight pits or bins for the reception of manure, and these have been adopted in many places, particularly in California. The regulations of the District of Columbia specify that all manure must be kept in “covered receptacles,” and it has been found that a tight-covered barrel suffices for a one-horse stable. Some workers in the Federal Department of Agriculture have carried out a series of experiments indicating that borax, hellebore, and calcium cyanide are highly destructive to maggots in horse manure, and it is quite probable that some very cheap and simple treatment will soon be developed.

The proper disposal of human excreta is another problem that is vitally connected with the checking of germ-laden flies. As Howard says, “the average person in the large city has no idea of the fact that there are many comparatively intelligent citizens who in sanitary matters have not even reached the grade of civilization which demands the sanitary privy. Stiles, in the course of his great work in the Southern States, has brought together some startling figures. He is responsible for the statement that of 4,825 farm houses in six different states 2,664, or fifty-five per cent, have no privies of any kind; of 2,499 houses inhabited by white people, thirty-five and three-tenths per cent have absolutely none, and of 2,326 inhabited by negroes seventy-six and eight-tenths per cent have none. And what shall be said of the condition of a large part—the very great majority—of those which do exist? The uncared-for privy is still a most important factor all over the United States, even in portions of our most cleanly cities.” Herrick points out that “there is no longer any excuse for the old open box privy, cleaned out once a year. It is a menace to every house in the vicinity as well as to individuals living perhaps hundreds of miles away because of its possibilities in contaminating milk. Some form of sanitary closet must be substituted.” One of the very best is that designed by Stiles, and described in Farmer’s Bulletin 463, U. S. Department of Agriculture. The bulletin is free, and the privy can be built at a cost of five or six dollars.

Another fly which looks almost exactly like the ordinary housefly is *Stomoxys calcitrans*, usually known as the biting housefly, but sometimes called the stable fly because of its great abundance about horses and cattle. The stable fly may be recognized by the fact that it holds its head higher than the housefly, but Howard says that “the best way to distinguish between the two flies is to allow them to walk over your hand; if it bites it is

Stomoxys; if it does not it is probably the housefly.” The housefly has no mouthparts adapted for biting, but the stable fly is provided with a sharp awl-like proboscis, and sucks the blood from its victims. It is a great lover of direct sunshine, and, when not engaged about cattle and horses may be seen sunning itself on walls and fences. In damp weather it invades houses and verandas, and it is then that the silk-clad ankles suffer. This habit has given rise to the popular idea that houseflies bite just before a storm. The bite is painful at the moment, but there is no subsequent swelling or irritation; the puncture, according to Howard, does not seem to be poisonous to man, and aside from the pain given it is far less dangerous than a mosquito bite.

The life history of *Stomoxys* has been worked out by Newstead, who found that the creamy-white eggs are laid about sixty at a time in irregular clusters, usually in manure of some kind, but frequently in piles of decaying grass or straw. Howard reared the fly from cow manure, and says that the egg usually hatches in two or three days. “The larva need not be described,” he says, “because it is so similar to that of the housefly.” Newstead found that in this stage they lived from fourteen to twenty-one days, but that the absence of excessive moisture and the admission of a little light materially retarded development, which then extended over a period of thirty-one to seventy-eight days. In the puparium the insect remained from nine to thirteen days. The development of the species is therefore much slower than that of the true housefly. It is Newstead’s opinion that the winter is passed chiefly in the pupal condition.

The stable fly is considered by Austern and others to distribute various diseases among cattle, but is probably not dangerous to man. Brues remarks that it was once “thought to be a carrier of poliomyelitis (infantile paralysis) but it now seems probable that such is not the case.” Howard says, “I judge from the fact that it is attracted to human excreta that it may become a carrier of intestinal diseases,” but it has never been proved to be more than an accidental agent in this matter.

Bishop reports a great outbreak of this pest in Texas, where they were found breeding in straw stacks, and caused serious injury to cattle and horses. A similar case recorded by Lucian Iches of Santa Fe, Argentina, is described by Howard: “The biting flies swarmed on a large estate in almost incredible numbers. The cattle were driven nearly crazy by them. Certain

valuable Durham bulls which were observed were covered with the flies. They had lost their hair in large spots and the skin was cracking. Monsieur Iches naturally sought at once for the principal breeding places of the flies, and found them to be in the stacks of debris from the threshing of wheat and flax. Larvae and puparia were found by the millions in the lower portions of these piles of straw, where some fermentation had already begun. The sensible measure which he recommended was to have this debris burnt within forty-eight hours after the completion of the threshing, the ashes being used for fertilizing purposes. It turned out that there was an old provincial law in the province of Santa Fe ordering the burning of all debris after threshing, but it had not been carried out during recent years, and therefore the *Stomoxys* multiplied until this veritable plague ensued.”

Another plague of stable flies occurred some years ago in Oregon, but the life history was not known there and the breeding places were never found. The severity of the pest’s bite is remarked by Osborne: “It causes a great deal of annoyance to horses, cattle and other domestic animals, and is frequently very troublesome to people working in places where it abounds: Its bite is not poisonous and aside from the pain given and the possibility of its disseminating disease, it is less injurious than some other members of the group. When abundant, however, the annoyance may be very great.” Bold describes the pitiable condition of cattle bitten by the stable fly: “In some of the severe cases the joints were so much swollen that the poor animals could not bend their legs to lie down; and in them the inflammation rose so high as to cause the loss of the outer skin and hair.”

The cluster fly (*Pollenia rudis*) is a little larger than the housefly, and appears longer because the wings overlap in such a fashion as to make the body appear very slender. The thorax bears many short, yellow hairs, and the grayish abdomen inclines to be iridescent. During the Summer this fly lives upon flowers and fruits, and does not come near houses, but the adults seek sheltered places in the Fall, and spend the Winter in large groups or clusters—hence the name. When the clustering process happens to take place within a human habitation the flies become a nuisance. Howard quotes one of his correspondents from Illinois: “They seem to prefer to occupy the rooms on the north side of the house and those that are used but little. They gather in large bunches in the corners and along the edges of the ceiling. They cannot be driven out like other flies, but must be killed

outright to get rid of them, and when you mash them the odor is like that of honey. We have tried nearly everything that was recommended to us, with no effect. It seems impossible to get rid of them, or to keep them out of the house, for they crawl in through the smallest places in the windows.”

They enter one by one through small cracks or crevices, and, as one of Herrick’s correspondents writes, “screens and other devices which work perfectly in excluding the ordinary fly are useless in keeping these out.” W. H. Dall, describing the pest at Geneva, New York, states that “people soon learned to look everywhere; in beds, in pillow-slips, under table covers, behind pictures, in wardrobes, nestled in bonnets and hats, under the edge of carpets, etc.”

We are singularly ignorant of the life history of this insect. Macquart and Desvoidy have found *Pollenia* larvae in manure, and the Bureau of Entomology people reared a single specimen from cow dung. Keilin, however, claims to have discovered the maggot living as a parasite in certain species of earth-worms. It is said that the adults are particularly susceptible to the same fungous diseases which attack the housefly.

The lesser housefly (*Homalomyia canicularis*) is generally regarded by laymen as an immature specimen of the common housefly, but is really a distinct species, belonging to a different family altogether. The housefly, or any other fly for that matter, does not grow at all in the winged state—the newly emerged adult is quite as large as it will ever be. The *Homalomyia* maggot is very different from that of the ordinary housefly; it measures only about one-fifth of an inch in length, and is covered with spiny processes, to which the dirt clings and gives the creature a mottled, dirty appearance. They develop in various kinds of garbage and manure, being especially partial to human excrement. Hewitt has found them very abundant in privies, and Herrick remarks that “this habit of breeding in excreta of various kinds makes the flies dangerous inhabitants of our rooms. They may act as conveyers of disease germs quite as readily as the housefly. They are rapid breeders, for a generation may be produced in two weeks in hot weather.”

The blue-bottle fly (*Calliphora erythrocephala*) has a dull-colored thorax and a dark metallic blue abdomen. It is sometimes known as the blow-fly, and lays its eggs on meat and dead animals, or even in sores upon living

creatures. A single female has been known to lay more than six hundred eggs, and Hewitt says that twenty-three days is sufficient to produce a generation. The blow-fly ordinarily lives in the open air, but occasionally gets into houses, particularly in stormy weather. Herrick remarks that “it has been found frequenting human feces and for this reason may be suspected of bearing intestinal bacteria, thus making it a fly to be dreaded.”

The so-called cheese fly (*Prophila casei*) is a little shiny black fellow which develops in cheese, bacon, ham, chipped beef and the like. The larva is known as the cheese skipper because of its remarkable agility. Howard says that it will sometimes jump three or four inches, and I have myself seen these vigorous little maggots, stimulated by the heating of their home in a piece of Southern bacon, leap clear out of the frying pan into the camp fire. The cheese fly is not native to America and was probably imported in an early shipment of Swiss cheese. The insect is not particularly dangerous, but as Howard says “it is well to remember that not only has it been reared from dead bodies, but that it is also attracted to excreta of all kinds.”

The moth fly (*Psychoda minuta*) is a very fragile little insect whose broad wings are hairy like those of a moth. It is very common among weeds, clinging to the under surfaces of leaves, and often enters houses, passing easily through the ordinary fly screen. According to Herrick, “the larvae of some of these flies live in cow dung, others in decaying vegetation, while some live in water, especially sewage water or drain water from kitchens. We have often seen hundreds of these moth flies among the weeds overhanging a ditch carrying the drainage water from a kitchen. We have also seen them in abundance along ditches carrying sewage water from houses. In the first-mentioned instance they were always present on the window panes of the kitchen, readily passing through the ordinary wire screen. Judging them from the places in which they breed, we would consider them unwelcome guests in our houses.” The moth flies of Southern Europe bite human beings, and are carriers for a disease called phlebotomus fever, but the species so far identified in the United States probably do no particular damage.

The tsetse fly (*Glossina palpalis*), the sole distributor of the deadly sleeping sickness which has decimated the native population of equatorial Africa, is a near relative of our common housefly. The reproduction of the tsetse fly is very different from that of most insects, however. According to

Brues, “the female does not deposit her eggs, but a single one develops to the fully grown larval condition before being deposited. This larva soon pupates in the shade beneath the brush bordering the water where it has been dropped by the parent fly, and later emerges in the winged adult condition. The pupae requires such moist shade, and it is apparently for this reason alone that the flies never occur away from the immediate vicinity of the water. As a result of their method of development, the tsetse flies do not multiply rapidly, and even under favorable conditions only one larva is produced in a ten-day period.... It was thought at first that by moving all the natives back from the edges of the water, the flies thus left without opportunities for reinfection, would become free from trypanosomes (the organisms causing the disease) and that by isolating and treating cases of the disease in fly-free areas it would be possible to eliminate them entirely. In conjunction with this, the cutting of brush, especially about boat landings and watering places, has been practiced as far as possible. Contrary to expectations, it has been found that even after three or four years, infected flies still occur along the uninhabited shores. This led to experimentation upon animals and it is now known that various wild antelopes as well as certain domestic animals may act as reservoirs for the virus of sleeping sickness, which may thus persist in the complete absence of any human subjects. As a result of this discovery the great difficulties of combating the disease ... have been vastly increased.”

MOSQUITOES AND DISEASE

The mosquito is nothing more than a small, slender fly, very similar to the housefly in every essential detail of its structure. There are about sixty species and varieties in America, but the three most common and important species are: the ordinary house mosquito (*Culex pipiens*) which does not carry disease, the malarial mosquito (*Anopheles quadrimaculatus*) which disseminates malaria, and the tiger mosquito (*Aedes calopus*) which is a carrier of yellow fever.

The common house mosquito has perhaps no equal as a persistent and universally damned destroyer of human tranquility. Heartily detested everywhere, even by entomologists, no serious studies of the life history were made until comparatively recent years. The female deposits her eggs upon the surface of still water in boat-shaped masses, each mass containing from fifty to two hundred eggs. The floating egg-mass is always large enough to be seen with the naked eye, and often measures a quarter of an inch in length. The eggs usually hatch in two or three days, and the larva is the well-known wiggler which swims with a peculiar jerky motion in every rain-barrel and roadside pool in the country. The wiggler breathes through a tube near the caudal end of its body, and usually hangs head downward at the surface of the water, with the tip of the breathing tube in contact with the air. The food consists largely of minute organisms collected by two vibrating tufts of hair about the mouth. After a week or so of this wriggling life, the larva is transformed into another stage called the pupa, which has a larger head and thorax, and rides differently in the water because the breathing tubes have shifted to the upper part of the body. In five or six days more the skin of the pupa splits open and the adult mosquito slowly emerges. It usually rests quietly on the floating pupa-skin for a few moments, and then, as soon as its wings are sufficiently dried, flies away in search of food and mates.

The male *Culex* is much handsomer than the female, being provided with large plume-like feelers or antennae, by which he may be easily recognized. Another important sex-difference is that the *male doesn't bite*, but lives by

sucking the nectar of flowers, like a bee. The female of the species is deadlier than the male, even among mosquitoes, and it is the female that does the blood-sucking. The mosquito's beak consists of several sharp lances inside a sheath, the whole apparatus admirably adapted to pierce the human skin and transfer human blood to the stomach of the mosquito. Incidentally some poisonous substance, probably a salivary secretion, flows into the wound and produces more or less inflammation and swelling. The *Culex* or house mosquito, so far as is known, does not transmit any disease to human beings—it is detested only because of the pain which follows its bites, and its intolerable buzzing.

The malarial mosquito (*Anopheles quadrimaculatus*) deposits its eggs singly on the surface of the water, never in rafts like those of the house mosquito. The eggs are very seldom found in barrels or tubs about houses—the *Anopheles* prefers ditches, or creeks, or the shallows of slowly flowing rivers. The malarial larvae do not hang head downward at the surface as the *Culex* wigglers do, but lie flat just beneath the surface film, and may be identified by this characteristic horizontal position. The *Anopheles* pupa does not differ greatly in appearance from that of the house mosquito. The adult is somewhat larger than *Culex*, and the wings are gray-spotted instead of clear. Another important difference is this: *Culex* alights and rests with its body parallel to the surface which supports it, while *Anopheles* rests with its body at an angle—head down, tail up. Herrick says that he has seen them “clinging to the ceiling of a horse stall by their four front legs, with their bodies hanging almost straight downward.”

It has been known for a long time that malaria, otherwise known as ague, or chills-and-fever, was connected in some way with stagnant water. It was obvious enough to everybody that the disease was most prevalent in low, swampy regions, and most people regarded it as a result of poisonous gases which somehow arose from the surface of the waters. The word *malaria* means *bad air*. It was later discovered that the real cause of the disease is a microscopic bit of jelly-like protoplasm—a protozoan parasite which lives in the red cells of the blood. Brues tells us that “the protozoan blood parasites which cause malaria were first demonstrated many years ago, in 1880, by a French surgeon, Laveran, who discovered them in the blood of persons suffering from malaria. Five years later an Italian, Golgi, distinguished three kinds, each associated with one of the more familiar

types of malaria. They were found to go through a regular life cycle in the red blood corpuscles and, from analogy with other known protozoa, it was suspected that in addition to their non-sexual generations in the human blood there must be a sexual development in some cold-blooded animal. Manson was led to suspect that some insect might be the secondary host and, working on this hypothesis, Ross in India first found the malarial parasites in a certain kind of mosquito in 1898. He had worked for nearly three years with a common mosquito belonging to the genus *Culex* without result, but finally in a mosquito of the genus *Anopheles* was able to trace the development of the parasite. His epoch-making discovery has since been amply confirmed and extended by experimental proof until we now know that the various types of malarial blood parasites complete their life cycles in anopheline mosquitoes, the latter acting as the sole carriers of the disease.”

Malaria, then, is acquired in one way only, and that is through the bite of the malarial mosquito. A mosquito sucks a little blood from a malarial patient, this blood containing some of the protozoan germs which are the immediate cause of the disease. In the stomach of the mosquito the parasites reproduce sexually, penetrate the stomach walls, and proceed finally to the salivary glands, from which position they are injected into the blood of the next victim of the mosquito’s blood-lust. The parasites reproduce asexually in the man’s red blood corpuscles, and then burst out by millions into the liquid part of the blood. This happens simultaneously in all parts of the body, and the patient is shaken by the chills characteristic of the disease. Some varieties of the germs reproduce every two days, producing what is known as tertian fever; other types form spores every three days, and the resulting disturbance is known as quartan fever. After a while male and female germs appear in the patient’s blood plasma, but these never develop unless the sufferer is bitten by another mosquito. When this occurs, the sexual germs pass into the mosquito’s stomach, reproduce sexually, and are distributed as described above. Another man bitten, more non-sexual reproduction, more chills and fever, and so on *ad infinitum*.

A healthy person may, of course, be bitten by *Anopheles* and have a great many malarial germs in his body without developing malarial fever, as they sometimes lie dormant in the blood without reproducing at all. Ross has estimated that “something like a quarter of a billion of them must be present

to produce fever.” The more infected mosquitoes that bite one, the larger the number of germs that will be introduced, and the more severe the disease. The essential result of all the investigations is concisely stated by Herrick: “Malaria is caused by a minute animal parasite that lives within the red blood corpuscles of human beings. The parasite destroys millions of the red blood cells that are so necessary to life, and, in addition, secretes certain poisonous substances known as toxins, which lodge in various parts of the body.... Since this parasite lives only in man and the mosquito, it can get from one person to another only through the agency of these insects. In other words, a person once free from the malarial parasite will remain free just so long as the bites of certain species of mosquitoes can be avoided.” One acquires malaria in one way only, and that is by being bitten by *Anopheles* mosquitoes.

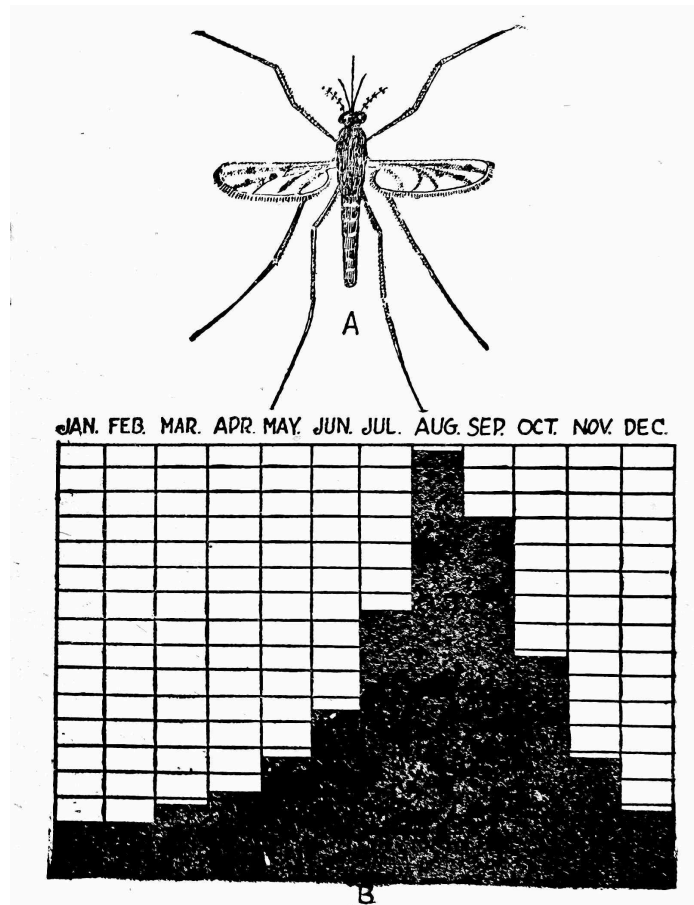


Fig. II. Malaria and its Carrier. A, the malarial mosquito, *Anopheles quadrimaculatus*. B, chart showing the seasonal prevalence of malaria in the United States.

The tiger mosquito (*Aedes calopus*) is not native to the United States, but has been introduced from the tropics. Today, however, it is widely distributed in the Southern States, and has been found as far north as New York. The eggs are laid singly or in small groups in stagnant water, and hatch in from twelve hours to three or four days. The larval life lasts about nine days, and the larvae are much more active than the other mosquito wigglers. The tiger mosquito is not often found in the country, but always near human dwellings, and it prefers to deposit its eggs about houses, in vessels containing only a little water—tubs, cisterns, rain barrels, old tin cans and the like. The pupa state lasts only about two days, and the whole development from egg to adult is sometimes passed in less than a fortnight. The adult is considerably smaller than the other mosquitoes described here,

and is conspicuous for its white banded legs, and the prominent white stripes on the thorax.

In 1881 a man named Carlos Finlay claimed that yellow fever was transmitted by mosquitoes, but nobody paid any serious attention to this novel theory, as everybody thought yellow fever was highly contagious—carried about in the clothing like smallpox. In 1900 a commission of United States army officers—Drs. Reed, Carroll, Lazear and Agramonte—went to Cuba to make an experimental study of the situation there. The procedure and results are concisely summarized by Herrick:

“In a field near Quemado, Cuba, this commission of surgeons erected a small wooden building tightly ceiled and with the windows and doors closely screened so that no mosquitoes could enter. In this house, during a total of sixty-three days, seven non-immune men were kept. These men slept in beds furnished with the unwashed pillow-slips, sheets, and blankets that had previously been used on the beds of genuine yellow fever patients in Havana and elsewhere. This bedding was actually stained with the excretions of the fever patients. Neither during that time nor subsequently did one of these seven men develop a case of yellow fever. This indicated to the surgeons, beyond much question, that yellow fever is not carried in clothing, as had always been held.

“Another house was built in this same field and divided into two rooms by means of a wire screen extending from floor to ceiling. The doors and windows of each room were closely screened with fine wire netting so that no mosquitoes could enter. All bedding and material carried into the rooms were disinfected by steam, which precluded any possibility of the yellow fever germ being present in the bedding or clothing.

“In one of the rooms, mosquitoes of a certain kind that had previously bitten patients sick with yellow fever were placed. In the other room none were allowed. Non-immune men were placed in both rooms. Of those in the room containing no mosquitoes, not one had yellow fever. Of those in the other room that were bitten by the infected mosquitoes, six out of seven developed cases of genuine yellow fever. This indicated beyond much question that mosquitoes were transmitters of this disease.

“These experiments have been extended and duplicated many times with the same results, so that we are justified in believing that a certain mosquito

known as *Aedes calopus* is the sole and only agent in the transmission of yellow fever.”

Dr. Lazear died from fever during the course of the experiments—a true martyr of medical science—and Dr. Carroll came very near losing his life also, following the bite of an experimentally infected mosquito. Among other things, it was discovered that a mosquito may bite a person suffering from yellow fever without becoming infected, unless the bite occurs during the first three or four days of illness—later than this the poison is not present in the blood. Another interesting fact is that the *Aedes* cannot infect anyone until at least twelve days after it has received the virus from a sufferer, but after this period it remains infected for a long time, and may give the disease to a large number of people.

The actual cause of yellow fever is as yet undiscovered. As Herrick says, “it is either too small to be seen with any lens now made or it inhabits some organs of the body not suspected, or its habits are entirely different from any other parasite with which we are familiar. In any case the germ has eluded all efforts to locate it and scientists are still ignorant regarding its real nature, habits, and appearance.” It has been supposed that the yellow fever organism is similar to the protozoan which produces malaria, but we do not know. Very little of importance has been learned about the disease or the mosquito since the Cuba commission in 1900, but wonderful progress has been made in the application of such information as we have. By 1902 Havana was entirely free from the yellow fever, and Rio de Janeiro, which had been a fever-hell for many years, eliminated the disease entirely after a six-year fight with the mosquitoes. In the Panama Canal Zone an army surgeon named Gorgas was very successful in combating both yellow fever and malaria. Although the United States has never suffered as the tropical countries have, there have been numerous epidemics, the last of which broke out in New Orleans in 1905. Vigorous anti-mosquito campaigns were waged and the plague stamped out in a short time. The tiger mosquito is still common all over the South, but there are no yellow fever patients for them to bite. When a case does appear—some sailor from the tropics, usually—he is seized by the Public Health Service people and placed where no mosquito can get at him, and thus it has not been necessary to wage a general exterminative war on the yellow fever mosquito.

When the situation gets out of hand vigorous measures for the temporary and local eradication of the insect are of course necessary. As Brues says: "The success of this campaign has undoubtedly sounded the death knell of the yellow fever epidemic and panic in the United States, for New Orleans has amply demonstrated what may be accomplished in the control of an epidemic by an efficient group of workers backed by a sympathetic public and supplied with reasonable funds. Even in parts of the tropics where it persists throughout the year, it is being rapidly and permanently eliminated. Indeed, it bids fair to be the first disease actually to become extinct as a direct result of human discovery and applied science."

There is, so far as is known, no satisfactory method of destroying mosquitoes in the adult state, so that in the war against this pest we must direct our efforts against the eggs and young. Herrick tells us that "the methods taken to destroy mosquitoes fall into three distinct classes; namely, the draining of bodies of water liable to contain eggs and wrigglers, the application of oil to bodies of water that cannot be drained and the introduction of fish into pools that cannot be drained or oiled."

Probably the best way to rid one's immediate vicinity of mosquitoes is to drain all marshes, pools and open receptacles. The drying up of a few acres of swamp land may cause a marked fall in the death-rate of an entire neighborhood, particularly in regions where malaria is prevalent. The exposure of open tanks, reservoirs and barrels in which stagnant water stands is no less than criminal in some parts of the South.

Very often, for economic reasons, it is not practical to drain a swamp or to eliminate open sewage ditches. In such cases we may take advantage of the fact that the larvae of mosquitoes do not breathe by gills, but are forced to come to the top for air. If ordinary kerosene oil is sprayed upon the surface of the water, it spreads to form a thin film which the larva cannot penetrate with their breathing tubes, and so are drowned. The kerosene also destroys the floating eggs, and kills the female mosquitoes as fast as they attempt to deposit. Crude petroleum has been used instead of kerosene, but it does not spread as well, and is very little cheaper. According to Herrick, one ounce of kerosene is enough for fifteen square feet of water surface, and half a teacupful is plenty for a barrel of water. The most convenient way to apply the oil to small pools and ditches is by means of a five-gallon tank carried knapsack fashion, with a short hose and nozzle. As the entire

development of the mosquito from egg to laying female seldom requires more than eighteen or twenty days, it is best to use the oil once every two weeks during the mosquito season. It would seem that an easier way to get kerosene into the sewage ditches would be simply to pour it into the water closet, and this has proven successful in cases where there is little or no current; on the whole, however, the surface-spraying system is best.

Ornamental ponds and pools cannot be drained or sprayed with oil, and the same may be said of artificial ponds where domestic animals drink. In these cases the mosquito pest may be mitigated by introducing fish or other small creatures which feed upon the larvae. In Japan, and more recently in other countries, the goldfish has been found very satisfactory for this purpose. Various species of shiners, sunfish and top minnows have also been used. W. P. Seal, who has given a great deal of attention to this phase of the mosquito problem, states his results as follows: "The writer has come to the conclusion, after many experiments in small ponds, that a combination of the goldfish, which is ornamental and useful in the open water, the roach or shiner, which is a very active species, two small species of sunfish, which live among plants, and the top minnow would probably prove to be more effective in preventing mosquitoes breeding than any other fishes."

Vernon Kellogg, Tillyard and others have studied the habits of dragonflies in this connection, and the latter particularly has emphasized their value as destroyers of mosquito larvae. He saw a single dragonfly larva eat sixty wigglers in ten minutes, and captured an adult dragonfly which had over a hundred mosquitoes in its mouth—so many that the mouth could not be closed. "I believe that a successful checking of the mosquito pest in the ornamental waters of parks and gardens could be readily obtained by the introduction of dragonflies whose larvae, as well as the adults, would prey upon the nuisance."

The use of wire screens to exclude flies and mosquitoes from dwellings is now well-nigh universal in most parts of the United States, although there are still many rural communities in the South in which it appears to be quite unknown. To keep out ordinary mosquitoes the wire should have at least fourteen meshes to the inch, and the yellow fever mosquito will pass through any screen which has less than eighteen meshes. It is practically impossible to fit screens and frames well enough to exclude all mosquitoes,

but it is certainly better to contend with a few stray mosquitoes in a screened house than to do battle with the myriads of pests which swarm into an unscreened domicile. The fewer the bites one receives, the fewer the chances of being bitten by an infected mosquito; the fewer infected bites, the less severe the resulting illness will be.

In many Southern cities the use of bed nets of bobbinet or mosquito bar has now become quite general. There is usually a kind of wire frame over the bed, and the net is sometimes tucked under the mattress all round, while in other cases it is long enough to reach the floor on all sides. Many travelers who must visit small towns, where the hotel bed nets are lacking or unsatisfactory, carry their own nets, and set them up every night to suit their fancy. Dr. Ross says: "Perhaps our first and best defense against malaria lies in the habitual and scrupulous use of mosquito nets at night.... The first care of the resident in the tropics, of the traveler, the sportsman, the soldier, the miner, the clerk, should be for his mosquito net. Wherever he lives, wherever he goes, he should see that his mosquito net is with him, that it is in good order, and that it is properly arranged at bedtime."

There are several chemical substances used in driving mosquitoes out of houses. One may rid a room of mosquitoes by the fumes of burning sulphur, but the gas often discolors articles painted with lead paint, and tarnishes gilt furniture and brass bedsteads. A vegetable powder sold as pyrethrum, buhach, or Persian insect powder is made of chrysanthemums, and is fairly effective when scattered about or blown into cracks and crevices. It may also be burned as a smudge, but the fumes are not as effective as those of sulphur. Many of the mosquitoes fall to the floor in a stupor, and must be gathered up and burned, as otherwise they will revive in an hour or so. Another fumigant much in favor in New Orleans is a mixture of carbolic acid and gum camphor, and is known as culicide. A heavy reddish liquid, it is evaporated by heating slightly over a gas burner or alcohol lamp. The mixture is inflammable but not explosive, and the fumes are not dangerous to human life.

Hunters and campers often smear their hands and faces with various substances supposed to discourage mosquitoes, oil of citronella being one of the most popular. Camphor, cedar oil, kerosene and mixtures of these substances in various proportions are also used, besides several patented smudges and ointments. I have tried several of these, but found them

unsatisfactory. Tobacco-smoke helps a little sometimes, and is at least free from the unpleasant and even nauseating odors of the other mosquito repellants.

LIFE AMONG THE BEDBUGS

The bedbug (*Cimex lectularius*) has been known to man for a long time, and has probably, as Herrick remarks, “been man’s bedfellow as long as man has slept in beds,” if not longer. The ancient Romans were intimately acquainted with this insect, and Pliny recommends a mixture of macerated bedbugs in water as a cure for snake bites. It is common in all parts of the civilized world, and doubtless came to America with our sturdy fathers—even the Mayflower probably carried a full cargo of bedbugs. Like its relative the stink bug, *Cimex* has a peculiar odor which is quite noticeable in small rooms where the insects are unusually abundant. The full-grown bedbug is a flat-bodied, mahogany-colored creature, with mouthparts admirably adapted to blood-sucking. It lives in beds, window casings, cracks in floors, etc., or in any article of furniture which affords a crevice large enough to contain it. What we know about the insect’s life history is mainly due to Marlatt, who did his work in 1896, and to Girault, who carried out some further investigations some ten years later. The eggs are very small, white and oval, and are, according to Herrick, “laid in batches of varying numbers in cracks and crevices in the bedsteads or other places where the bedbugs happen to be. The number of eggs deposited by a single female is not known. Southall, Riley, and others have made the common statement, probably not based on actual observation, that each female lays about four batches of fifty each during the season. Girault actually succeeded in obtaining 111 eggs from one well-fed female between June 17 and August 19. How many she had deposited previous to confinement for the experiment he was, of course, unable to say. Girault’s experience with this one bug indicates that the females may continue to lay eggs at different periods throughout the breeding season and that there is only one generation a year.”

The eggs hatch in about ten days, and the young bugs appear very much like their parents—bedbugs do not pass through larva and pupa stages like flies and mosquitoes. Bedbugs probably eat nothing but blood, and a young individual, if properly supplied with this food, reaches maturity in about six weeks, shedding their skins at least five times during the period of growth.

It is said that the bug feeds but once between moults; if this is true the occupant of the bed must suffer at least five bites in order to rear a single bedbug to maturity. Bugs very much like *Cimex* are found upon swallows, chimney swifts, pigeons and martins, and many people believe that genuine bedbugs are distributed by these birds, but this is probably not the case. True bedbugs have never been found upon them, and although the “bird bedbugs” are occasionally found in houses they do not seem to enter beds and bite human beings. Bedbugs have, however, been found in chicken houses, where they presumably bite chickens, so that there is nothing inherently unreasonable in the view that they may be distributed by birds.

The bedbug almost invariably lies hidden during the day, or as long as there is a bright light in the room. As soon as the light is turned off they sally forth to attack an exposed portion of the sleeper’s body. It seems that they are particularly likely to be secreted about the pillow, and as soon as the room is dark hit upon the neck as the most desirable and accessible part of the victim’s anatomy. Only one who has had personal experience in these matters can appreciate the vigor of the onslaught, and the astounding rapidity with which the enemy whisks out of sight the moment the light is flashed on. The bedbug’s bite is not poisonous, and the pain is probably due only to the puncture, but this is sufficient to cause considerable irritation and swelling.

Elie Metschnikoff, the author of that fascinating book entitled *The Nature of Man*, was the first to call attention to the bedbug’s possibilities as a carrier of disease germs. Bedbugs have been made, under laboratory conditions, to transmit typhoid and other fevers, and a tropical disease called *kala-azar*, but it has never been demonstrated that the bugs actually do, under ordinary conditions, play any part in the diffusion of these diseases. “Many writers and experimenters,” writes Herrick, “have labored hard to prove this insect guilty of graver offenses than that of simply stealing blood from human hosts. The most they have been able to do so far, however, is to show that in one case, at least, the bite of the bedbug formed a starting point for a case of bubonic plague. As a matter of fact, this is really a stronger indictment against the bedbug than, at first thought, might appear. The sores resulting from bedbug bites offer ideal points of entrance for disease-producing organisms and are a source of real danger. Actual and definite proof that the bedbug transmits disease is very difficult to obtain,

but suspicion points strongly in that direction.... It is extremely desirable to avoid the bites of this insect if possible, especially in hotels where beds are occupied by so many different people; but this is very hard to do, in fact, almost impossible if one travels much.”

Luckily the bedbug is wingless, and therefore much easier to control than winged pests like the fly and mosquito. If *Cimex* were able to fly there would be no keeping him out, as the newly hatched young could pass through practically any sort of screen. The use of metal bedsteads puts the bugs at a disadvantage because they afford fewer cracks and crevices than those built of wood. Boiling water poured into the crevices kills both eggs and bugs, but a more convenient method is to use an oil can or squirt-gun filled with gasoline, kerosene, turpentine or alcohol. The easiest way to rid a room completely of insects is to fumigate with hydrocyanic acid gas, made by pouring dilute sulphuric acid on potassium cyanide. This gas kills every living creature in the room, but it is dangerous in the hands of careless or thoughtless persons.

LICE, CRABS AND COOTIES

Lice have been associated with human beings as far back as our historical records go, and are mentioned in the writings of Aristotle, Herodotus and other ancient writers. Ordinarily cleanly and well-dressed people today would be horrified to find lice about their persons, as they are now confined to soldiers and others who do not keep their bodies and clothing clean, but a few hundred years ago, when everybody was dirty, they thought nothing of it. Samuel Pepys, an English diarist of the seventeenth century, made the following laconic entry in 1668: "Up betimes, finding our beds good, but lousy, which did make us merry," and there are many other matter-of-fact references to these vermin in the literature of the period.

There are three common species, but they are all essentially alike in structure, being small, wingless and provided with piercing and sucking mouthparts. Schiödte describes the behavior of a louse on the back of his hand: "Scarcely does the abominable little monster feel the heat of the skin before it lays aside its former disheartened attitude, and begins to feel at ease, its antennae oscillate for joy, and it stretches all six legs complacently out from the body. But though the pleasure and surprise at the sudden transportation into congenial surroundings for the first moment eclipse everything else, hunger soon asserts its claim, sharpened as it is by the long fast, which has rendered its stomach and intestines quite transparent. The animal raises itself on its legs, walks a few steps, seeking and feeling its way with its antennae, while we follow it with a magnifier. Presently it stops, draws in its legs a little, arches its back, bends the head down toward the skin at an oblique angle, while it probes a small dark and narrow organ repeatedly forward, and draws it back through the fore end of the head; at last it stands still, with the point of the head firmly abutted against the skin."

All three of our species are blood-suckers, and are strictly human parasites, being found only occasionally upon the bodies of animals other than man. It is said, however, that they are sometimes carried about by flies.

The head louse (*Pediculus capitis*) is usually found among the hairs on the head, occurring elsewhere only occasionally. The female is about one-twelfth of an inch long, while the male is only about one-half as large; both sexes are grayish in color, but it is said that this varies with the color of the host. Herrick quotes Murray to the effect that lice on West Africans are black, those on Chinese yellow, those on Hindoos smoky brown, and so on. The eggs are fastened to the hairs by a gelatinous secretion, and the female has been known to deposit at least fifty eggs in the course of a week. The period of incubation is usually five or six days, and the young louse reaches maturity in a fortnight. The movements and bites of these creatures are irritating in the extreme, and the sufferer is sure to scratch so incessantly that the scalp is always in an inflamed condition. The head louse is not known, however, to carry any disease-producing organism.

The best way to get rid of the head louse, according to a circular memorandum prepared under the direction of Surgeon General M. W. Ireland, of the United States army, is to “clip the hair of the head with a hair clipper and wash with a mixture of equal parts of kerosene and vinegar. This should be followed in a few hours with a bath of soap and hot water.... The hair should be caught in bags and burned. In order to reduce the liability to infestation, the hair should be kept close at all times.” Sulphur ointment may be used without sacrificing the hair, but it is not so effective as the kerosene mixture.

The body louse (*Pediculus corporis*), known also as the gray-back and the cootie, is somewhat larger than the head louse, and is usually of a dirty white color. These creatures dwell in the seams and folds of clothing, and are therefore known to some writers as *Pediculus vestimenti*. They are difficult to find and to dislodge, as they do not often cling long to the skin, but retire to their hiding places in the clothing as soon as their blood-sucking is done. The eggs are laid in the seams and wrinkles of the clothing, that made of wool being given the preference. When one has no opportunity to change or wash one's clothes the pests become so abundant as to be quite unbearable. Soldiers, inmates of prison camps and such people are particularly liable to be abundantly supplied with body lice; they may pass from one person to another when the bodies are brought into actual contact, or one may become infested by sleeping in louse-ridden beds.

The Dutch scientist Leeuwenhoek, more than two hundred years ago, carried out some experiments which are pleasantly reviewed by Herrick: "Leeuwenhoek ... made an attempt to find out something definite about the life history and rate of development of the body louse. He did not believe the popular saying that a louse could become a grandfather in twenty-four hours. At first he thought of hiring some person to act as host for the lice. Later he changed his mind, overcame his own natural aversion to these pests, and enclosed two large females within a fine black stocking, the top of which he fastened tightly around his leg above the knee. Here he allowed the two lice to live for six days and obtain their sustenance from his leg. At the end of this period he removed the stocking and found fifty eggs around one of the females and forty eggs in another part of the stocking, evidently laid by the second female, which, however, had escaped. He wore the stocking for yet ten days, when on examination he found twenty-five young lice which so disgusted him and dampened his enthusiasm that he threw the whole thing into the street. Since Leeuwenhoek's time the author is not aware that any scientist has ever tried in the same way to study the life history of these lice."

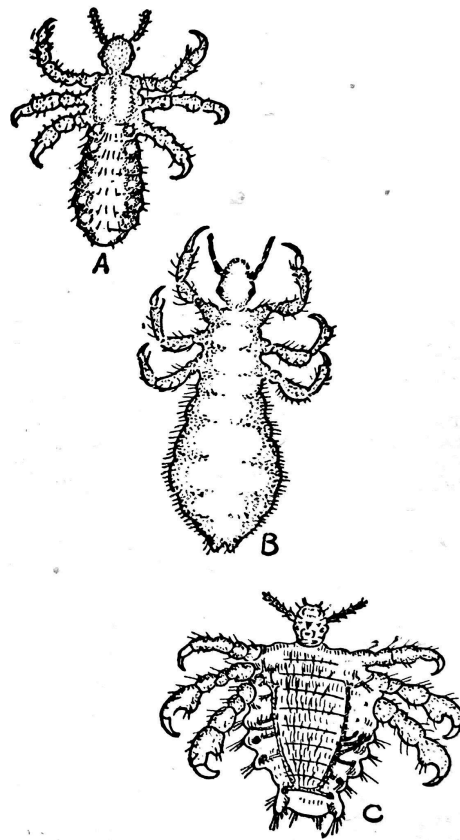


Fig. III. Three kinds of Lice. A, the head louse, *Pediculus capitis*; B, the body louse, gray-back, or cootie (*Pediculus corporis*); C. the crab or crab louse, *Pediculus pubis*.

Pediculus corporis is the only species of louse known to act as a carrier of disease. Its connection with typhus and trench fever is admirably described by Brues: "Typhus fever has been well-known for many years and regarded as a disease characteristic of filthy surroundings. During our own civil war it claimed many victims among the inmates of army prisons, and has been endemic though not very prevalent in many parts of the world in times of peace. Through the researches of Ricketts and others we now know that typhus is spread by the body louse and its epidemiology is at once made clear. When it broke out in Serbia in severe epidemic form, a

knowledge of the method of its transmission made control possible, even under extremely difficult and unfavorable circumstances.

“Trench fever has attracted notice in the European war zones, to which it appears to be restricted so far as present knowledge extends. That it is a new disease is, however, without question an utterly unwarranted assumption, for it has undoubtedly been brought to Europe from some little-known quarter of the globe, unless it may have previously existed in Europe which does not appear probable. During the latter part of the war it was successively recognized as a distinct disease, suspected of association with the louse, and soon proved actually to be louse-borne. We now know that the disease is due to a living microorganism, probably of such small size that it cannot be recognized under the microscope. This virus is obtained by the lice with their meal of blood taken from an infected person. At least five days must elapse before the louse becomes capable of transmitting the disease, indicating that the organism must undergo a development of definite periodicity in the insect. If it is transferred to another person its bite is not or only rarely infectious, but its excrement contains the virus and if scratched into the skin, trench fever develops. Typhus fever is a very dangerous disease with a quite high death-rate, but trench fever is non-fatal and its importance in the war zones has been due to its great prevalence and the fact that persons afflicted with it are often incapacitated for long periods.”

As body lice do not cling to the body except when actually biting, it is a comparatively easy matter to get rid of them. One has only to bathe while his clothing is being steamed or soaked in gasoline. The United States Army set up delousing plants which were nothing but steam sterilizers on a large scale, and they proved very satisfactory.

The crab louse (*Pediculus pubis*) is short and stout-legged, whitish on the back, with a distinct reddish cast about the legs. It usually inhabits the hairy areas about the sex organs, clinging close against the skin, but is not uncommon in the armpits, and has been found upon practically every part of the body except the head. The eggs are attached to the hairs, like those of the head louse; they hatch in less than a week, and the young crab reproduces at an early age. This little pest multiplies much more rapidly than either of the other species, and is besides more easily transmitted from one person to another. Although the usual means of communication is direct

bodily contact, as in the sexual embrace, it may be transmitted otherwise. The United States Army memorandum informs us that “this insect is transmitted mainly by contact in lodging houses, houses of prostitution, bathtubs, and perhaps occasionally from toilet seats.... Although the crab louse has not been shown to be a transmitter of disease, still it is very annoying, and its presence is a reflection on a man’s cleanliness.... The treatment for this condition is to shave the hair of the pubic region, axillae, chest and legs. This should be followed by an application of the kerosene and vinegar mixture, followed by a bath with soap and warm water.” The usual treatment among civilians is the application of a little mercurial or blue ointment.

OUR FRIEND THE COCKROACH

There are four common species of cockroaches in the United States, only one of which is native to this country. The American cockroach (*Periplaneta americana*) is the large winged species so common in the Middle and Southwestern states. They devour nearly any sort of food, and have been known to eat the corks out of bottles, while no less an authority than Vernon Kellogg says that some sailors in San Francisco sleep with gloves on to keep roaches from gnawing off their finger-nails. The German roach (*Ectobia germanica*), known also as the croton-bug, is common in the eastern part of the United States. It is the smallest species we have, being seldom more than five-eighths of an inch in length. For some reason or other these roaches are usually found about water pipes, and are not common in small communities which have no water systems. The Oriental cockroach (*Blatta orientalis*) is common also in England, and came to America with the early colonists, although it is supposed to have originated in Asia. It is larger and stouter than the croton-bug, and is dark brown or black in color. The males have short wings, but the females are wingless or practically so. The Australian roach (*Periplaneta australasiae*) is very much like the American variety, except for some yellow spots and bands on the thorax and shoulders. It is less common than any of the others, being found chiefly in the South.

The life histories of all four species are essentially identical. The eggs are not deposited singly, but in clusters enclosed in a horny capsule, the end of which may often be seen projecting from the abdomen of the female. The young roaches appear very much like their parents, but do not reach maturity for several months. The flat bodies of these insects allow them to crawl into small cracks in woodwork and furniture, and they usually remain hidden during the day, coming out at night to seek what they may devour. They are really scavengers, and under certain conditions may be useful in this capacity, but their omnivorous habits and the intolerable stench of their bodies and excreta make them the bane of the careful housewife.

“Cockroaches,” observes the indefatigable Herrick, “are among the most difficult to control of the household pests. They are difficult to reach because they are especially adapted with their thin, flat bodies for hiding away in inaccessible cracks, crevices and crannies. Moreover, they are wary and shy of all baits and traps. The croton-bug is the most difficult of all to get rid of. It seems to display more caution in avoiding traps and baits than most of the others, and as it increases faster, it becomes much more abundant.” Several traps have been devised and used with some measure of success, and roaches may often be killed by feeding them plaster of paris mixed with flour, or by exposing deep jars of stale beer, a delicacy in which they hasten to drown themselves. Herrick quotes Washburn to the effect that powdered borax scattered about the kitchen will clear a house of cockroaches. The best way to completely rid a house of these pests is to fumigate with hydrocyanic acid gas, made by pouring dilute sulphuric acid on potassium cyanide. But this method, as I have pointed out elsewhere in this booklet, is too dangerous for awkward or absent-minded people to experiment with.

CHIGGERS, TICKS AND FLEAS

The chigger (*Leptus irritans*) is not a true insect, but the immature form of a mite, related to the spiders. The eggs are laid upon the ground in the Spring, and when the young chigger hatches it climbs up a blade of grass and awaits the coming of some animal to which it can attach itself. At this time it is a very small red creature, barely large enough to be seen with the unaided eye. It clings to any moving object which presents itself—usually some passing insect, but quite frequently a bird or one of the higher animals. After having fed for a time upon the blood and juices of its mount, the chigger drops off and changes into the adult mite, which is not parasitic, but feeds largely upon plant lice.

Man's contact with the chigger usually comes while walking through the grass and weeds to which they cling. They cannot bear direct sunlight, and are not common upon closely clipped lawns unless these are very densely shaded. The tiny tormenter passes through the finest of clothing—silk hosiery and underwear are no protection—and attaches itself to the skin, sucking blood and setting up a severe irritation by means of some poisonous secretion. The resultant itching is so intense that one cannot resist the temptation to scratch the affected parts, which may become infected and cause blood poisoning, and, according to Hamilton, erysipelas. Although the chigger is not known to disseminate any particular disease, it is an intolerable nuisance, and has spoiled many a vacation in the South.

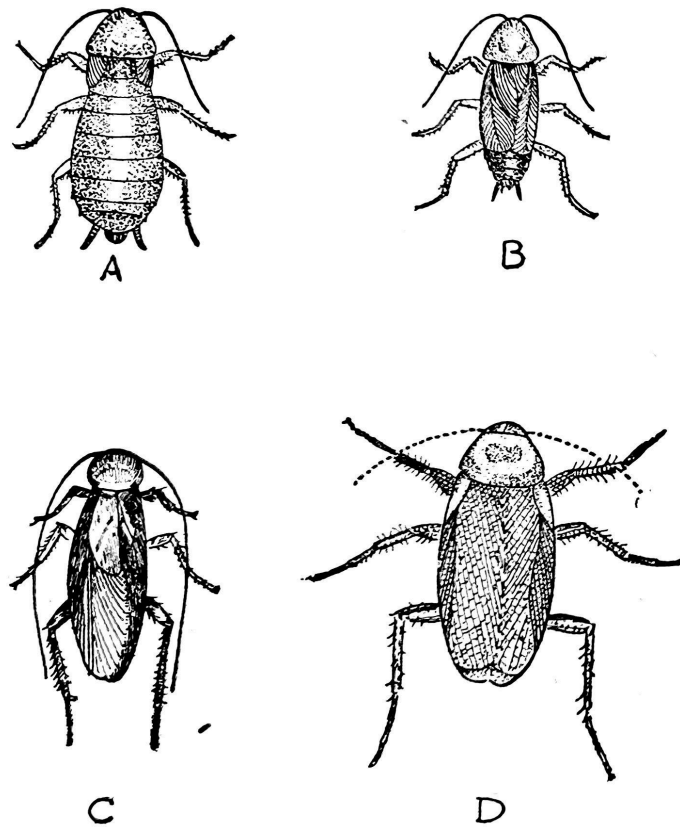


Fig. IV. Some Common Cockroaches. A, Oriental cockroach (*Blatta orientalis*), female. B, same, male. C, American cockroach (*Periplaneta americana*). D, Australian cockroach (*Periplaneta australasiae*).

Some persons avoid chiggers by wearing high boots, others, sprinkle sulphur in their stockings, while still others use various patent preparations both before and after the act. My own experience is that if, directly after coming in from the field, one changes to chigger-free clothing and rubs whiskey or gasoline upon the parts likely to be affected, the number of bites will be greatly decreased. Dr. Vernon C. Allison treats the inflamed areas with potassium permanganate, but I have had no satisfaction in the use of this remedy.

The fever tick (*Dermacentor venustus*) is the carrier of the disease known as Rocky Mountain spotted fever. This disease is now restricted to the Northwestern states, most of the cases occurring in Idaho and Montana. The life history of the tick is much like that of the chigger. Hatched from eggs on the ground, the young seed tick mounts a tall blade of grass or a weed and fastens upon the first animal which comes along. Forcing the hooked sucking beak into the flesh of the host, it sucks blood until it is full grown. Then it falls to the ground and reproduces, dying soon after this duty is done.

Brues describes the fever tick situation as follows: “The *Dermacentor* ticks occur abundantly on various small wild mammals in the younger stages and as adults on domesticated animals, such as cattle, and from these become transferred to man. It has been experimentally shown that certain rodents are susceptible to the disease, and a tick thus infected in the nymphal stage can retain the disease organism until it becomes adult. It may then reach its human host through the medium of domesticated animals such as cattle. It appears that this is the ordinary way in which human cases have their origin, i. e., through the bite of adult ticks, although the newly hatched *seed ticks* derived from eggs laid by infected mother ticks are known to contain the organism also.

“Although Rocky Mountain fever is of minor importance at the present, it is feared that it may increase its range at any time, since other ticks of wider distribution are apparently capable of acting as carriers. Whether this may happen is by no means certain, however, and the vigorous measures already undertaken to reduce the abundance of ticks on domesticated animals will undoubtedly bear fruit in the gradual reduction of this locally much-dreaded disease.”

There are many other species of ticks in the United States, but their life histories are all very similar. Several of them are believed to be carriers for various diseases of cattle, but *Dermacentor venustus* is the only one known to distribute disease among human beings. The common Southern cattle-tick (*Margaropus annulatus*) is the sole carrier of the cattle disease known as Texas fever, which has caused a tremendous pecuniary loss in several of the Gulf States. The protozoan parasite which actually causes the trouble is called *Babesia bigemina*, and Brues refers to the tick’s “becoming infected during its period of engorgement when feeding on the blood of a diseased

animal and then transmitting the *Babesia* through its eggs to the young ticks of the next generation. These may feed on healthy animals the next season, conveying to them the parasites that have been handed down from the mother tick."

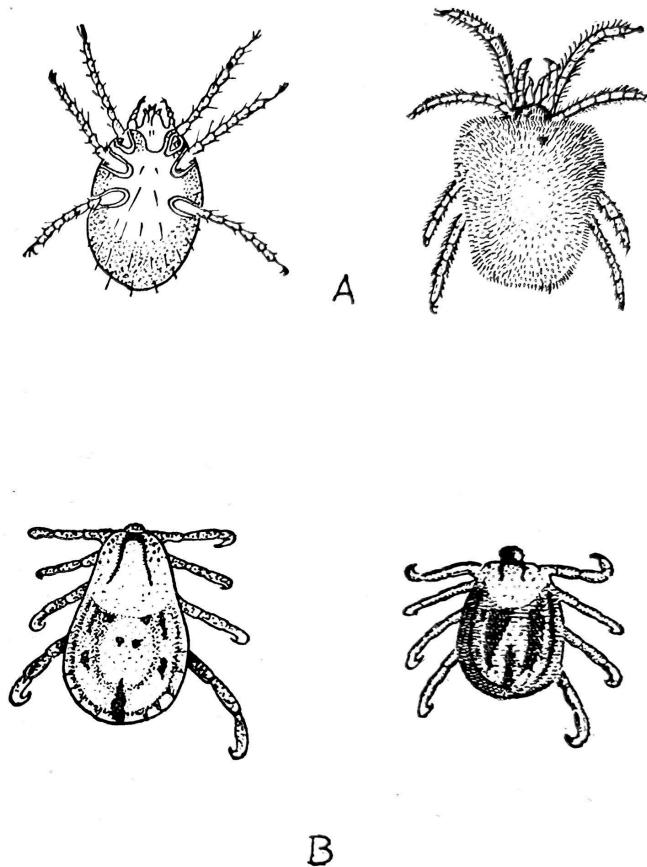


Fig. V. Chiggers and Ticks. A, young and adult chigger (*Leptus irritans*) redrawn from Herrick. B, Rocky Mountain spotted fever tick, (*Dermacentor venustus*), after Brues. Male at left, unengorged female at right.

Only two species of fleas are common in dwelling-houses in the United States. One is the human flea (*Pulex irritans*) and the other is *Ctenocephalus canis*, found almost universally upon cats and dogs. One must not suppose, however, that human fleas bite humans only, and that *Ctenocephalus* confines its attention to dogs and cats; the human flea does

not scruple on certain occasions of good omen to attack dogs, cats, rats and mice, while the various other species of fleas bite man also whenever the spirit moves them. The adult flea lives entirely upon the blood of its host, obtained through the admirable piercing and sucking mouthparts. The human flea, according to Herrick, is so bloodthirsty that it sucks regularly more blood than it can hold, and the surplus is seen squirting out of the anus while the creature is sucking more in at the mouth.

The eggs are dropped almost anywhere, and the tiny white larvae develop in cracks and crevices, feeding upon the organic matter in wood and rubbish. The pupa is enclosed in a silken cocoon, and the adult may appear in less than two weeks after the eggs are laid, although the period is usually somewhat longer.

Until comparatively recent years the flea was looked upon only as a nuisance, but it has now been discovered that some species act as agents in spreading bubonic plague, one of the most terrible scourges known to human history. It appears that the plague is primarily a rat-disease, and affects the human animal only incidentally. Outbreaks of plague among human beings are always preceded by an epidemic among the rats of the same region. The part played by fleas in the matter is stated by Brues as follows: "The relation of the flea to the transmission of plague is due to the fact that rats are regularly infested by fleas that may become infected with the bacillus of plague, if it be present in the blood of the host upon which they are feeding. These bacilli remain in a viable condition for some time in the gut of the flea and may be transferred to a human subject bitten by an infected flea. Thus, when a rat dies of plague, its fleas leave it to search for a new host; if they attach themselves to a rat, that animal is liable to infection, or if they feed upon a human being, as they frequently do, the disease may become transferred to man.... The plague bacilli (*Bacillus pestis*) appear only in fleas that have bitten infected persons or rats twelve to twenty-six hours previous to death, for after this time the bacilli do not occur in the blood. The vitality and virulence of the bacilli are preserved for nearly a week at least and sometimes fully a month; and there is actually an increase in their number during the first few days. Infection from these insects may then occur through their bites, if they contain extremely virulent bacilli, but probably occurs more commonly by the insects being crushed *in situ* after they have punctured the skin."

From 300,000 to 400,000 cases of bubonic plague occur in India every year, and nearly half of them terminate fatally. Comparatively few cases have been reported from the United States in recent years, but in various places, San Francisco for example, only the most vigorous and persistent repressive warfare against rats and fleas have prevented what might have been very serious outbreaks.

Transcriber's Note

Illustrations were moved to the ends of the paragraphs in which they originally appeared. Their original page numbers are in the list of illustrations. The list of illustrations was not present in the original text.

Inconsistencies in hyphenation have been standardized.

Minor punctuation errors have been changed without notice.

Spelling was retained as in the original except for the following changes:

Page 2 : “Our Friend the Cochroach”	to	“Our Friend the Cockroach”
Page 3 : “Europe. Picture Thomas Aquinus”	to	“Europe. Picture Thomas Aquinas”
Page 8 : “During the Spanish-American war”	to	“During the Spanish-American War”
Page 13 : “an epidemic which occured”	to	“an epidemic which occurred”
Page 13 : “contended that purulent ophthalmia”	to	“contended that purulent ophthalmia”
Page 14 : “is practically no ophthalmia”	to	“is practically no ophthalmia”
Page 14 : “spread of Egyptian ophthalmia”	to	“spread of Egyptian ophthalmia”
Page 16 : “destroyed by the housekeeper’s”	to	“destroyed by the housekeeper’s”
Page 20 : “walls and fences. in”	to	“walls and fences. In”
Page 20 : “and verandas, and is”	to	“and verandas, and it is”
Page 21 : “Lucian Iches of Sante”	to	“Lucian Iches of Santa”
Page 22 : “the province of Sante”	to	“the province of Santa”
Page 23 : “inclines to be iridescent”	to	“inclines to be iridescent”
Page 25 : “this habit of breeding”	to	“this habit of breeding”
Page 27 : “a single one developes”	to	“a single one develops”
Page 32 : “The parasites reproduce sexually”	to	“The parasites reproduce asexually”

According to other passages in the text, the parasite that causes malaria reproduces asexually in the human body and sexually in mosquitoes. This typo was corrected to make the text internally consistent.

Page 35 : “a field near Quemados”	to	“a field near Quemado”
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Page 37 : “is that the Aedes”	to	“is that the <i>Aedes</i> ”
Page 42 : “or Persian insect power”	to	“or Persian insect powder”
Page 47 : “writings of Aristotle, Heroditus”	to	“writings of Aristotle, Herodotus”
Page 47 : “ease, its antennae oscilate”	to	“ease, its antennae oscillate”
Page 48 : “antennae, while we followed”	to	“antennae, while we follow”
Page 49 : “is not known, howeve”	to	“is not known, however”
Page 54 : “application of the kerosense”	to	“application of the kerosene”
Page 60 : “cattle, but <i>Dermacentor venusta</i> ”	to	“cattle, but <i>Dermacentor venustus</i> ”
Page 63 : “years the flea looked”	to	“years the flea was looked”

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